

Identify Spatially Variable Genes using BSP and scBSP

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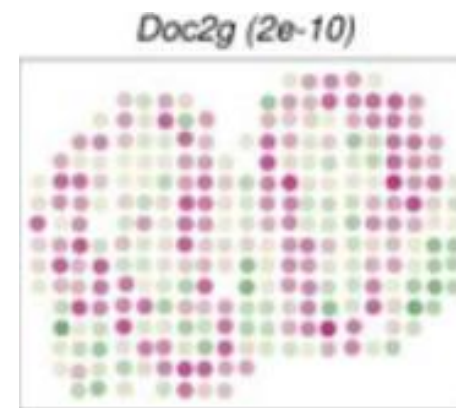
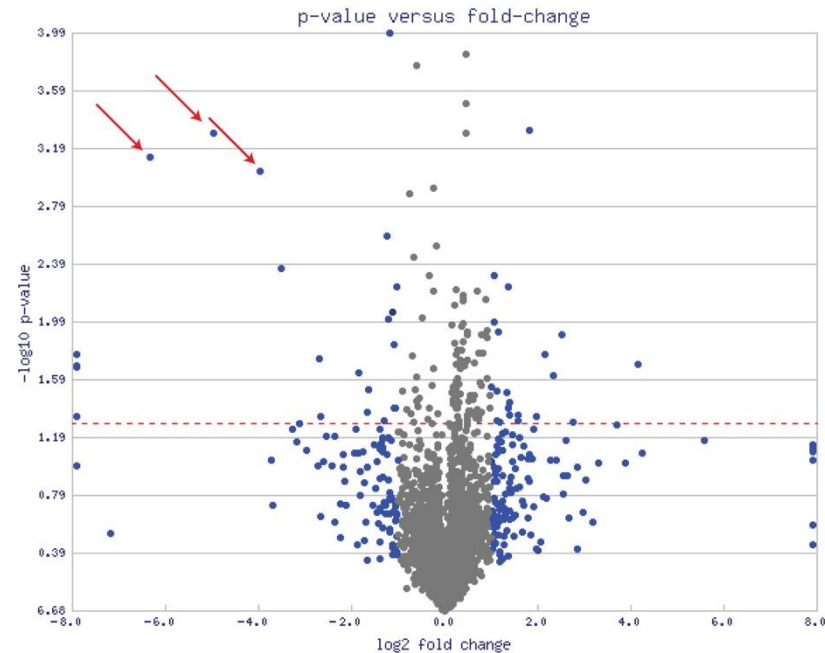
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07/07/2026



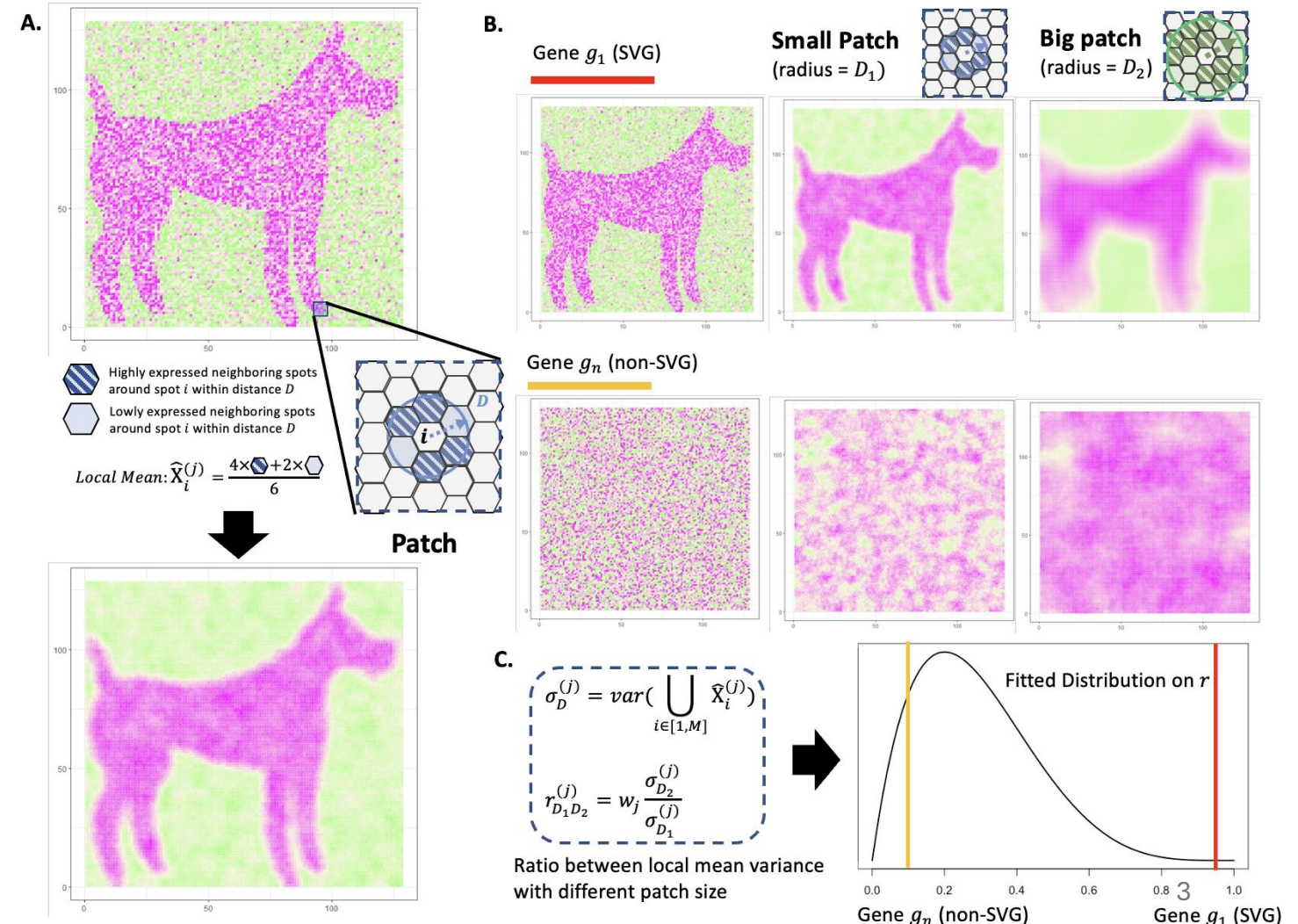
Spatially Variable Genes (SVGs) (spatial context of DEG)

- Differential Expressed Gene -> Spatial Variable Gene
- **Spatially Variable Gene:** Gene differential expressed because of spatial (condition as DEG)
- SVGs are defined as genes with a **highly spatially correlated pattern of expression**, which varies along with the spatial distribution of a tissue structure of interest.
- Spatial expression variation can reflect communication between adjacent cells, position-specific states, or cells that migrate to specific tissue locations to perform their functions.



BSP: Dimension-agnostic and granularity-based spatially variable gene identification

- BSP (big-small patch), a non-parametric model by comparing gene expression patterns at two spatial granularities to identify SVGs from two or three-dimensional spatial transcriptomics data in a fast and robust manner.
- **Velocity of change in variances of local means in different granularity can be used to distinguish SVGs and non-SVGs**
- Data driven, model free, any dimension
- Highlighted by Editor's focus in Biotechnology and methods



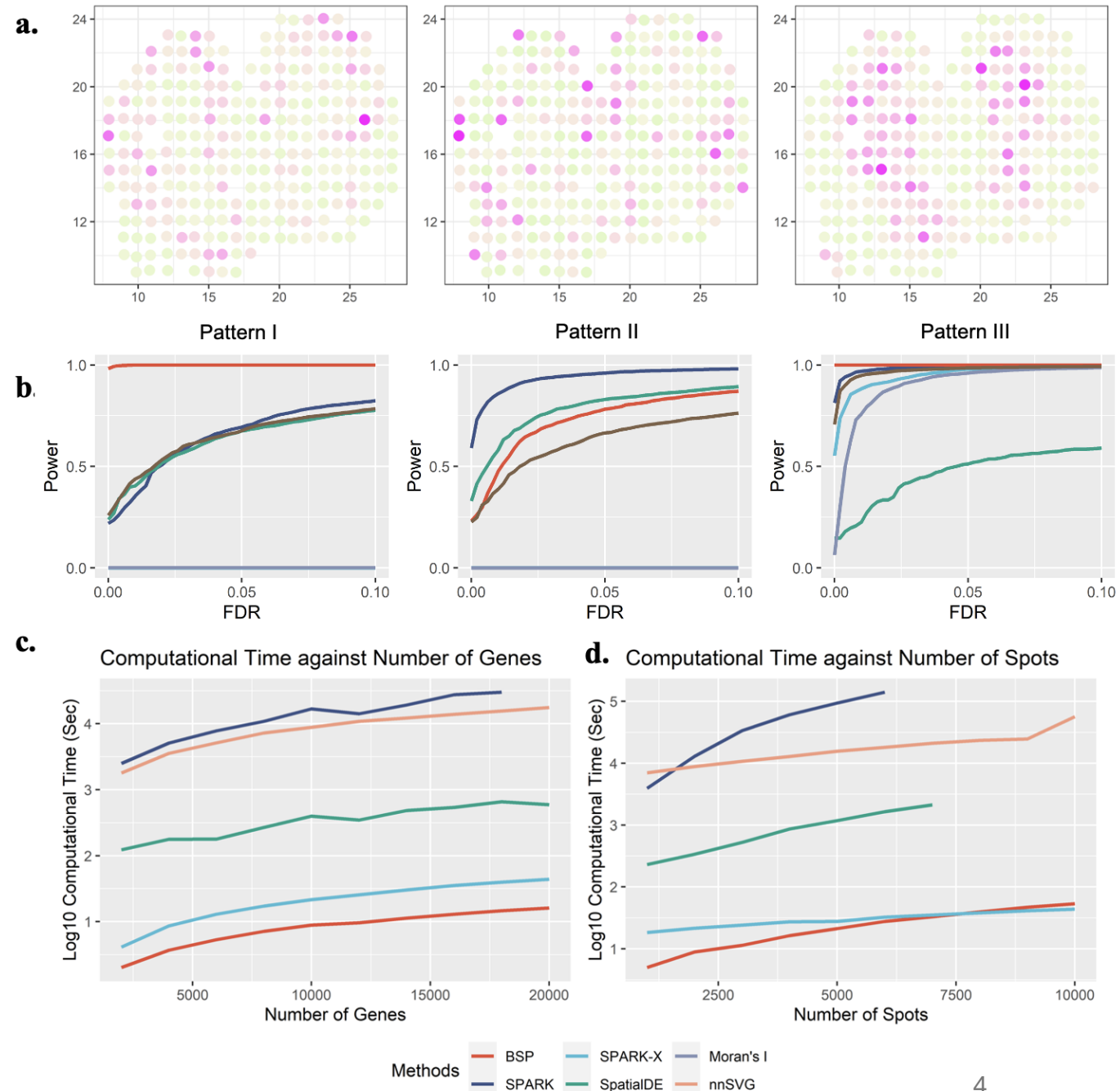
Wang, J., Li, J., *et al.* Dimension-agnostic and granularity-based spatially variable gene identification using BSP. *Nat Commun* **14**, 7367 (2023).

Li, J., Raina, M., *et al.* (2025). scBSP: a fast and accurate tool for identifying spatially variable features from high-resolution spatial omics data. *Bioinformatics*, 41(10), btaf554.

2D simulations

- Better performances on most scenarios varying spatial patterns, size of the pattern, signal strength, and noise level
- Fast and accurate

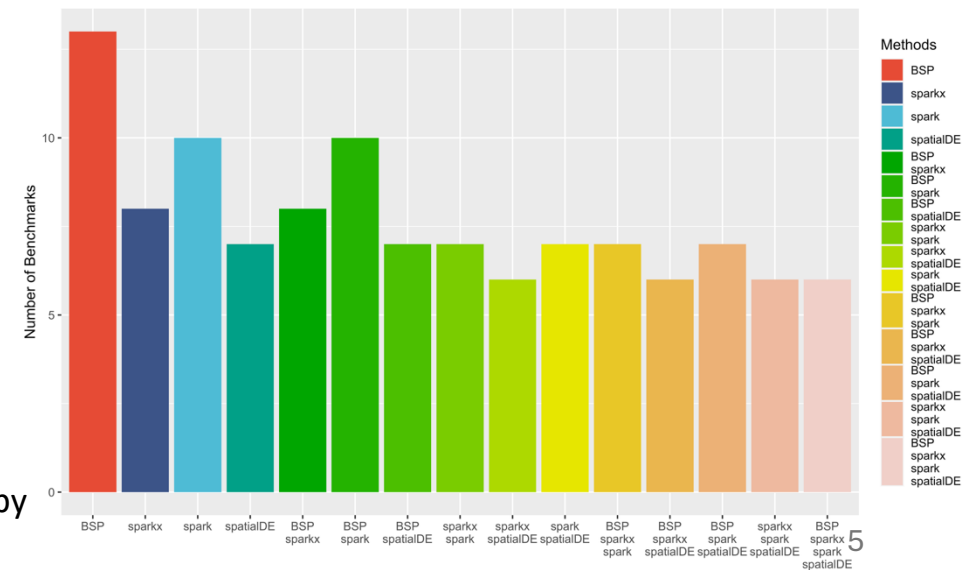
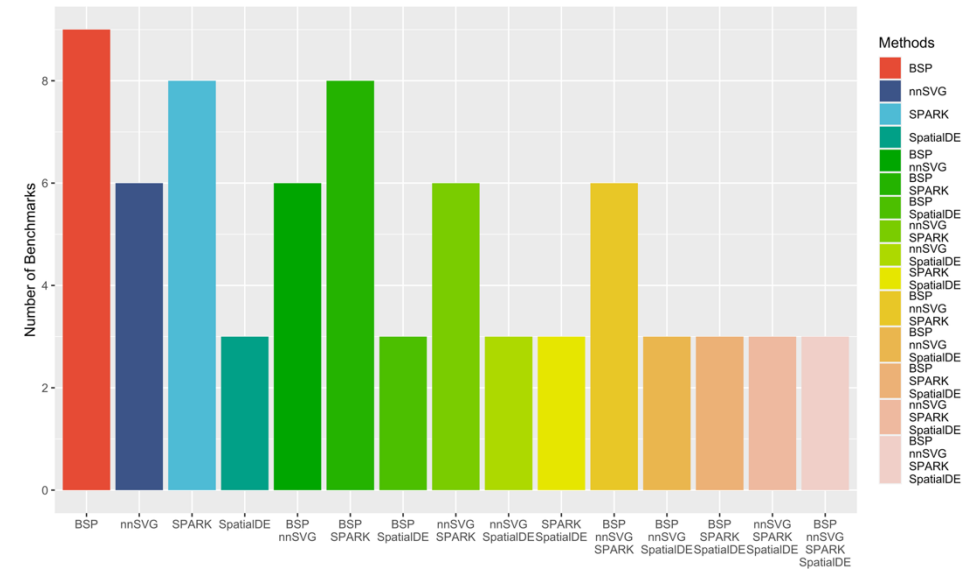
Mouse olfactory bulb (simulation)



2D Real datasets

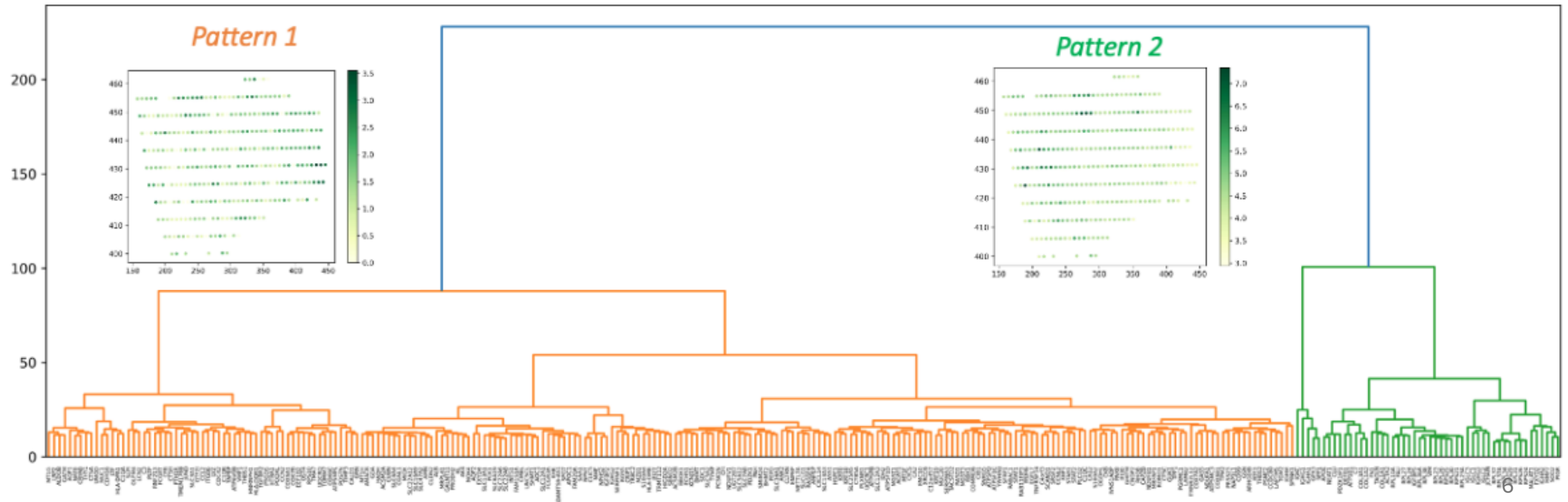
- Mouse factory bulb study
 - ST platform
 - 10 marker genes
 - BSP detected 9
 - SpatialDE detected 3, SPARK detected 8, nnSVG detected 6, and SPARK-X detected 0
- Human breast cancer
 - ST platform
 - 14 marker genes
 - BSP detected 13
 - SpatialDE detected 7, SPARK detected 10, and SPARK-X detected 8

Stahl, P. L. *et al.* Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78-82 (2016).

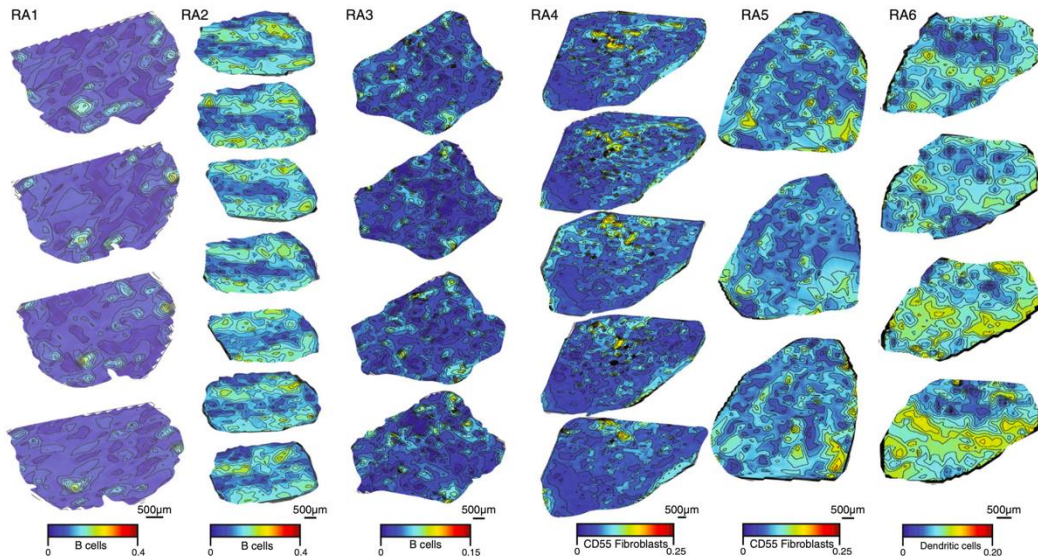
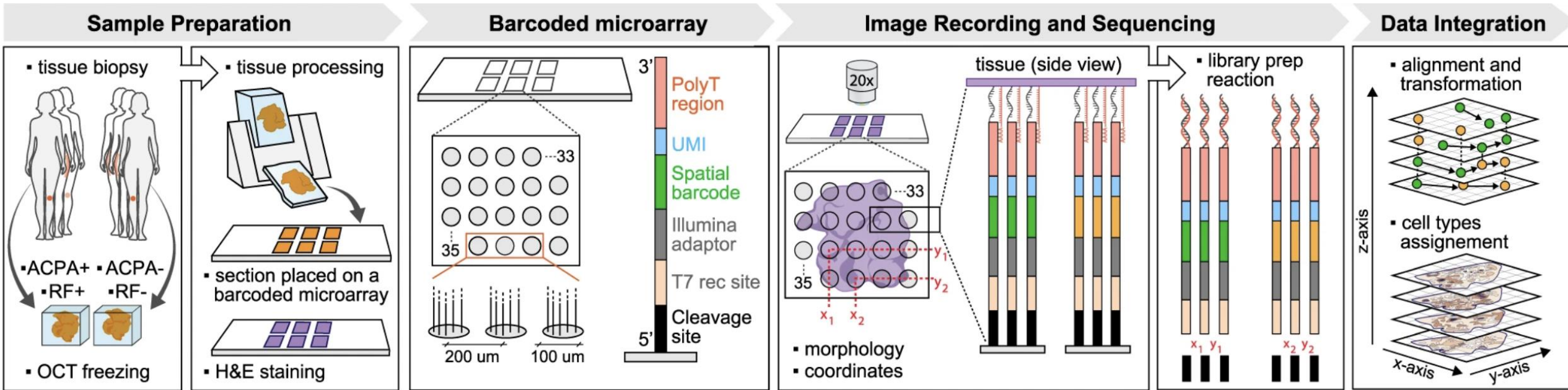


2D case study: Acute Kidney Injury (10X Visum)

- Pattern 1
 - Aerobic respiration (q-value $1.04\text{e-}09$) and oxidative phosphorylation (q-value $4.69\text{e-}09$), consistent with hypoxic or nephrotoxic acute tubular necrosis, the most common tubulointerstitial form of AKI.
 - Pathways of early recovery were enriched including **kidney development** (q-value $8.56\text{e-}07$) and **metanephric nephron epithelium development** (q-value $2.30\text{e-}06$), which included genes like PAX8.
- Pattern 2
 - Humoral immune response (q-value $5.15\text{e-}05$) and tissue homeostasis (q-value $5.15\text{e-}05$).
 - Several **immune responses** were activated (q-value $3.51\text{e-}04$) including B cell receptor signaling pathway (q-value $4.72\text{e-}07$).



3D spatial transcriptomics is on the way



Cut onions in slices

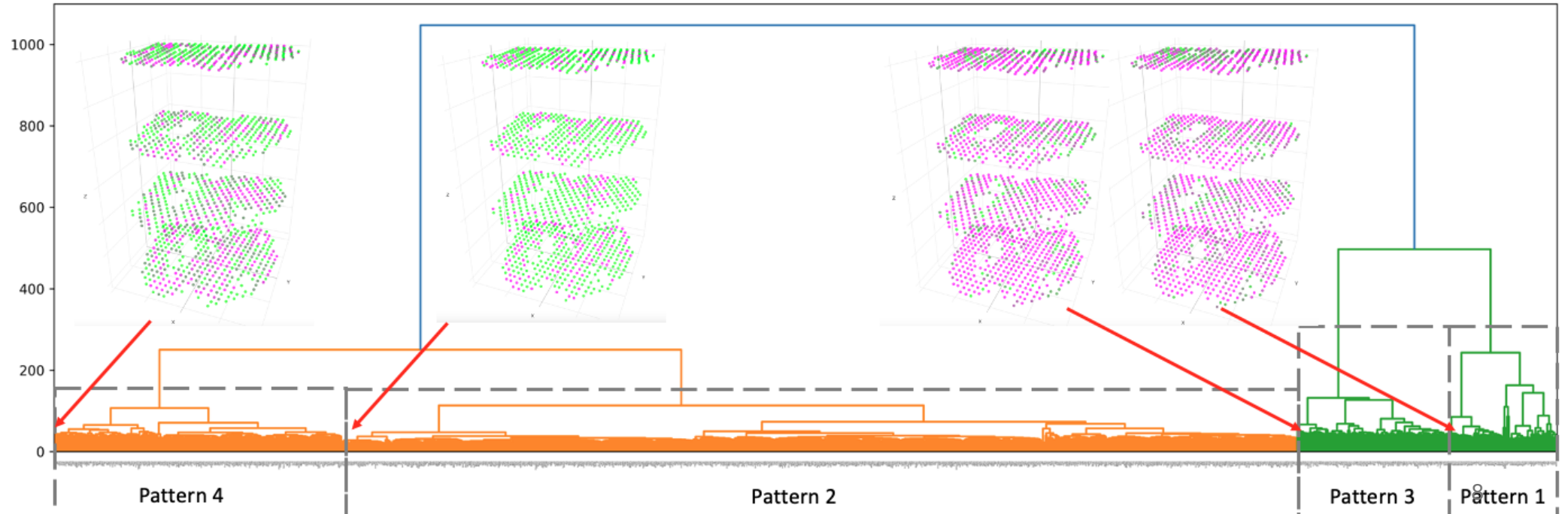
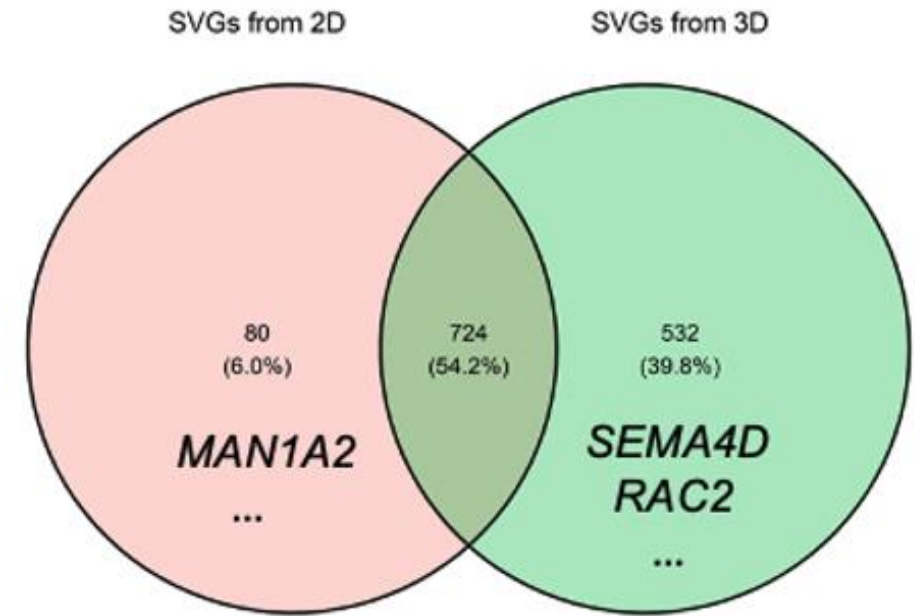
27 slices of 6 RA patients

Vickovic,..., **Aviv Regev** & Patrik L. Ståhl "Three-dimensional spatial transcriptomics uncovers cell type localizations in the human rheumatoid arthritis synovium." *Communications Biology* 5.1 (2022): 1-11. ⁷

3D case study: Rheumatoid Arthritis

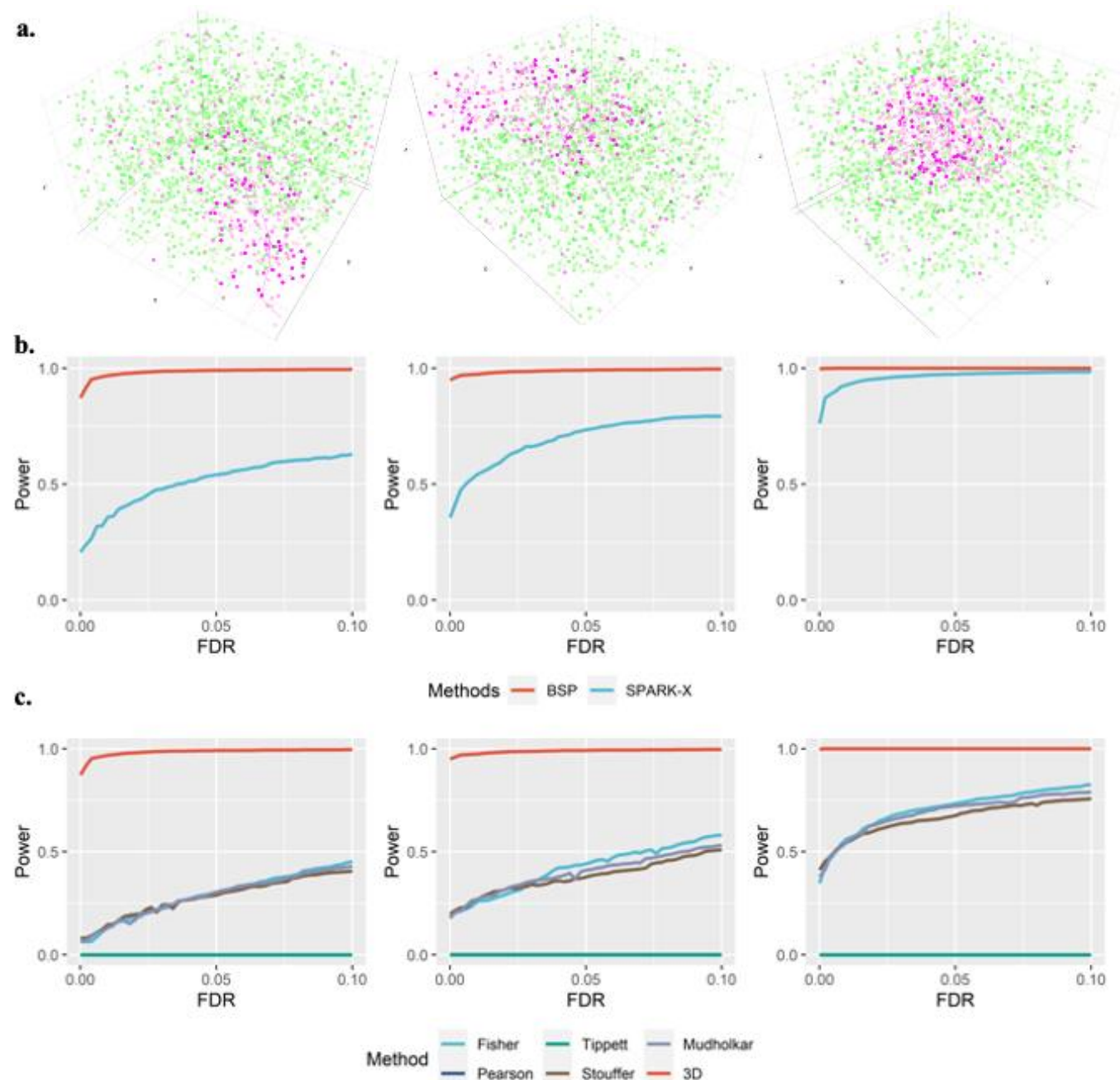
- Identified more biological meaningful genes than 2D meta-analysis

Vickovic, S. *et al.* Three-dimensional spatial transcriptomics uncovers cell type localizations in the human rheumatoid arthritis synovium. *Commun Biol* 5, 129 (2022)

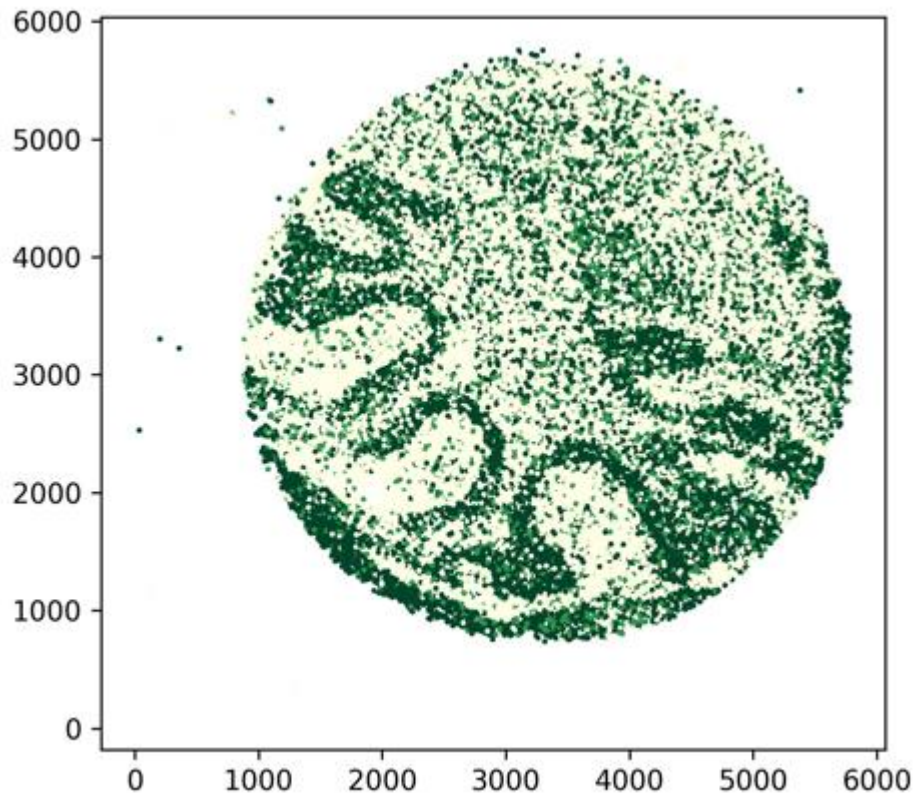


3D simulation

- Three 3D patterns:
- curved stick (Pattern I), thin plate (Pattern II), and irregular lump (Pattern III)
- Power analysis varying pattern size, signal strength, and noise.
- 3D outperforms meta-analysis on 2D slides

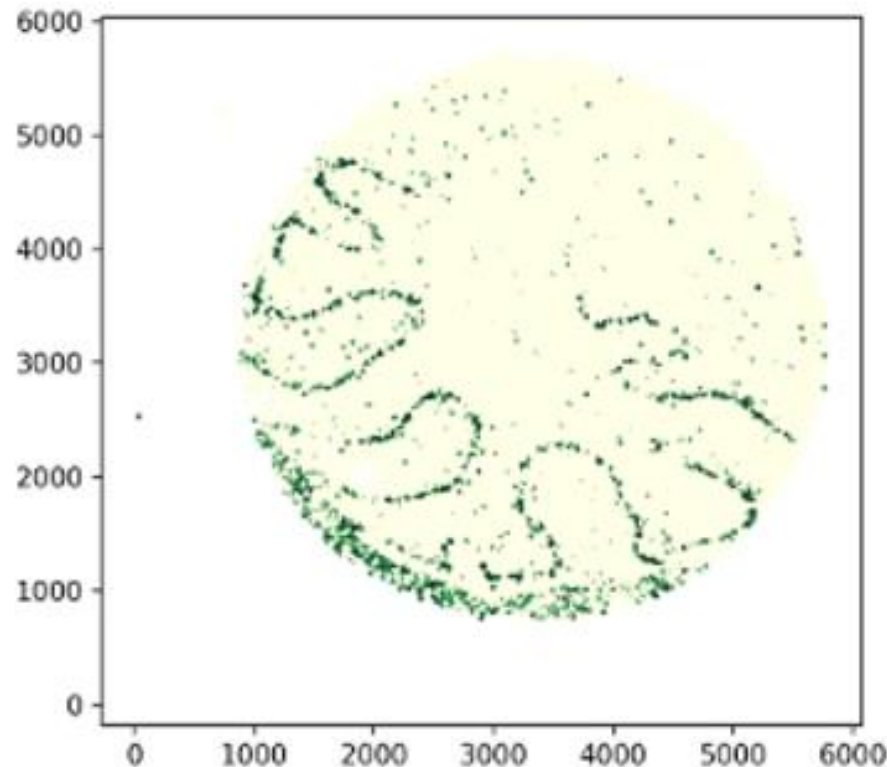


Challenges from large scale data



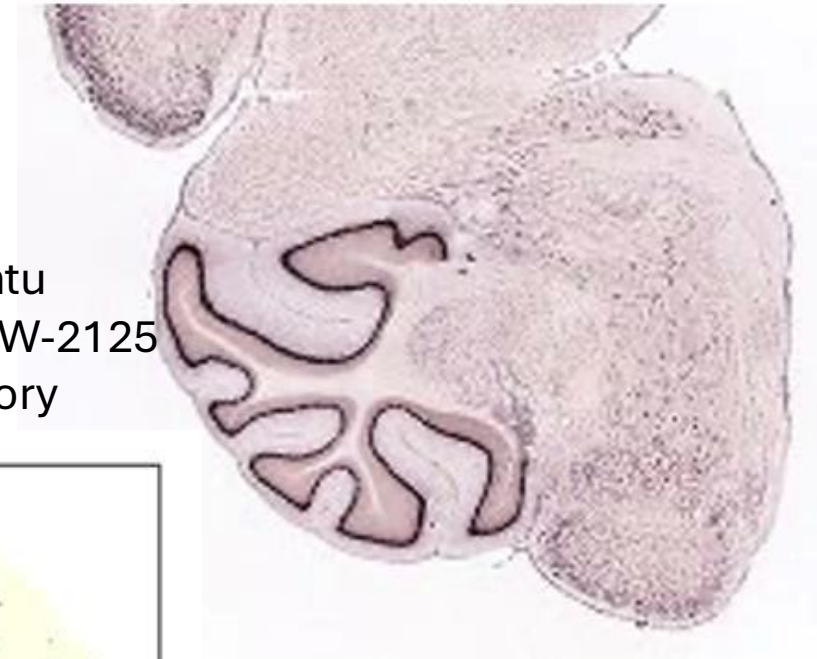
***Malat1* (left) and *Calb1* (right) as SVG identified by BSP**

23,096 genes and 39,496 beads
Running time: **18 min** on an Ubuntu workstation with Intel(R) Xeon(R) W-2125 CPU @ 4.00GHz and 32 GB memory



ISH of *Calb1* in an adult mouse brain from a separate study

2D slide-seq v2



scBSP (BSP 2.0): Improve computational efficiency on BSP

- Using sparse matrix operation to accelerate the computing.
- Targeting subcellular Xenium/CosMx/MERSCOPE data

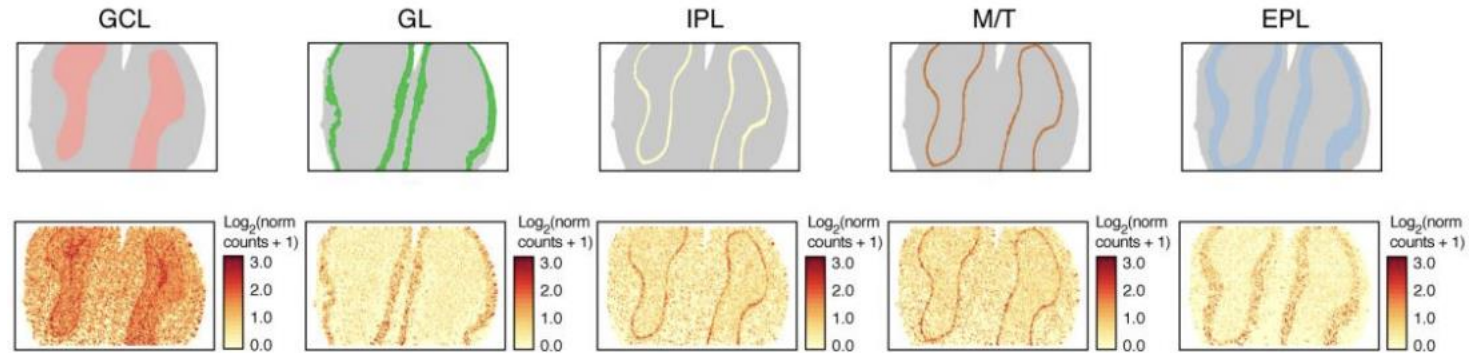
	BSP			scBSP		
Process	Method	Time Complexity	Space Complexity	Method	Time Complexity	Space Complexity
Data Pre-processing	Min-max Scaling	$O(NM)$	$O(NM)$	Maximum Absolute Scaling	$O(M)$	$O(NM)$
Patch Determination	Exhaustive Search	$O(NM^2)$	$O(NM)$	Approximate Nearest Neighbor Search*	$O(NM \log M)$	$O(NM)$
Expression Matrix Calculation	Dense Matrix	$O(M)$	$O(NM)$	Sparse Matrix	$O(SM)$	$O(SNM)$
Local Expression Calculation	Loop Operation	$O(NM)$	$O(NM)$	Matrix Operation	$O(SNM)$	$O(SNM)$

* The Approximate Nearest Neighbor Search in Python is based on HNSW implementation.

Table 1: Time and space complexity of BSP and scBSP. M: number of spots; N: number of genes; S: data sparsity (proportion of non-zero values in the expression matrix).

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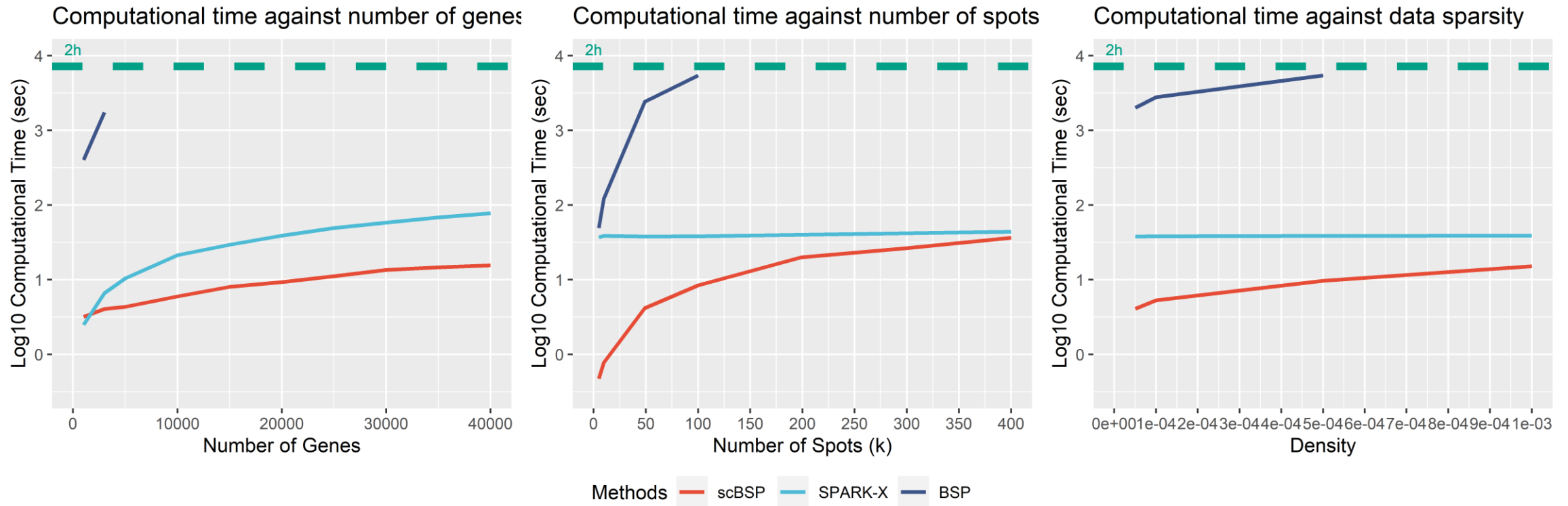
Performance on a laptop



- HDST data:
- ~200,000 spots, 20,000 genes, 0.0003 density
- BSP: 4 hours, 90GB
- scBSP: ~3 sec, 2GB

Vickovic, S., Eraslan, G., Salmén, F. *et al.* High-definition spatial transcriptomics for in situ tissue profiling. *Nat Methods* **16**, 987–990 (2019).

Results – running time



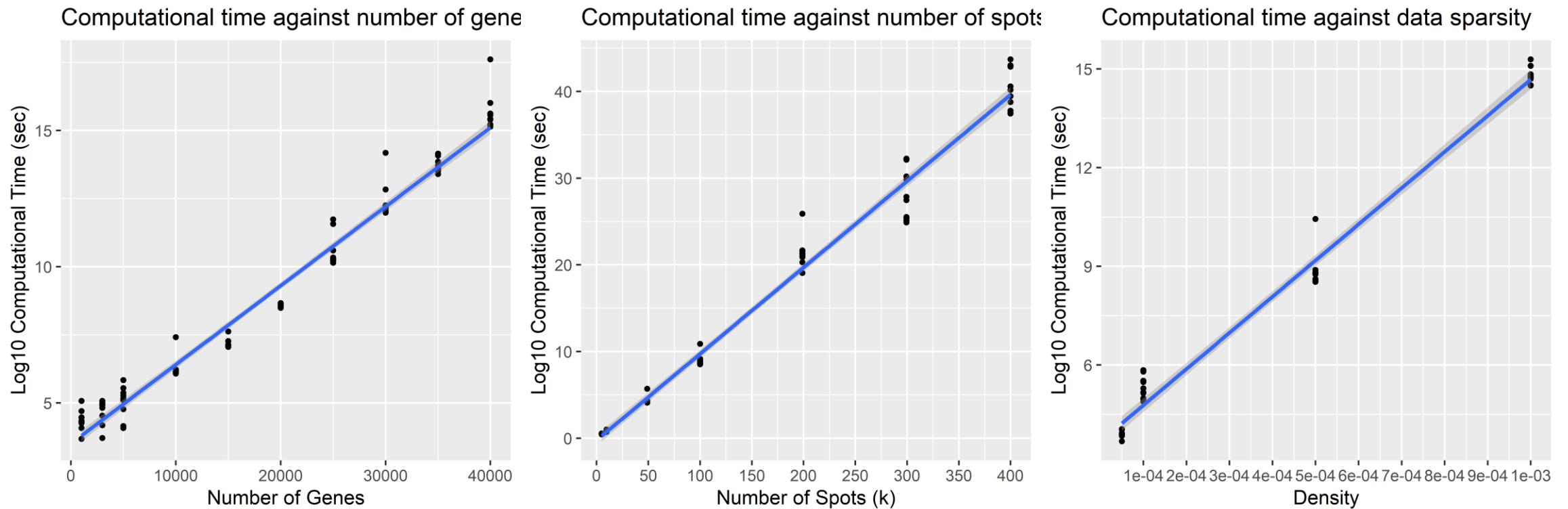
Left: Computational time with an increasing number of genes and fixed 100, 000 spots and density of 0.0005.

Middle: Computational time with an increasing number of spots and fixed 20, 000 genes and density of 0.0005.

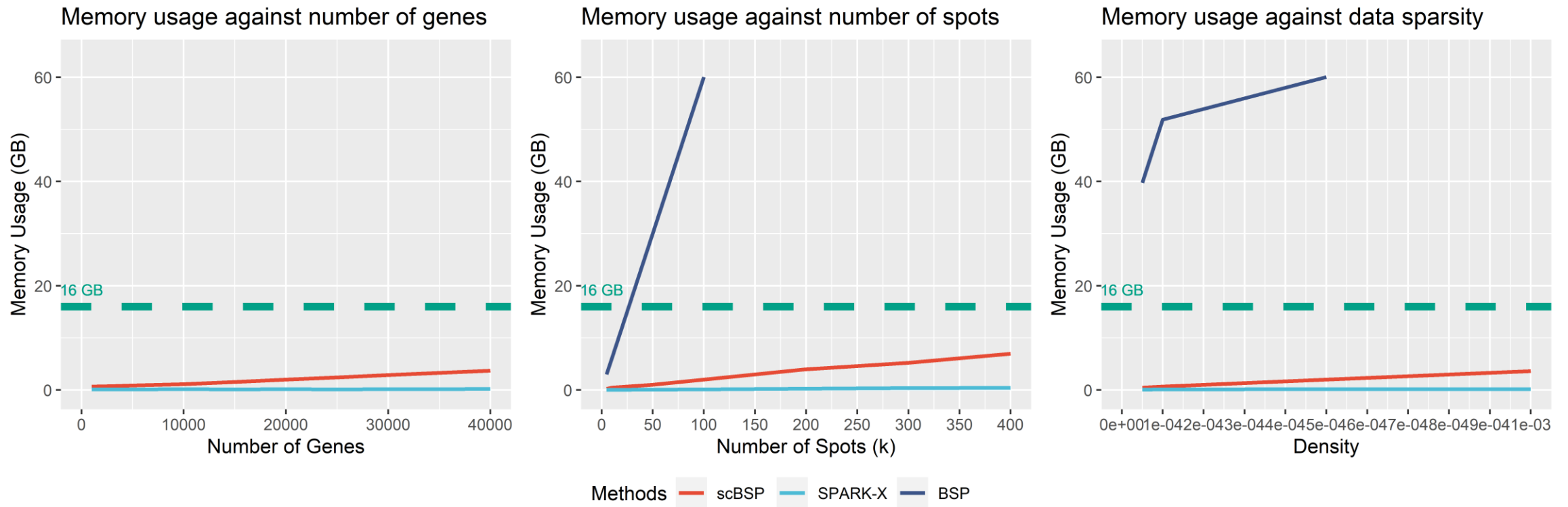
Right: Computational time with an increasing density and fixed 100, 000 spots and 20,000 genes.

Results – running time

Computational time scales linearly with number of genes, number of spots, and data density



Results – memory usage



Required RAM memory < 10 GB for most data

Acknowledgement

Lab Member

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ML and Data analysis

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Dr. Yuzhou Chang

Yi Jiang

...

Dr. Anjun Ma at OSU

Kidney Study

Dr. Michael Eadon at IU SOM

Dr. Krzysztof Kiryluk at Columbia

R: <https://cran.r-project.org/web/packages/scBSP/index.html>

Python: <https://pypi.org/project/scbsp/>

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